In Situ Raman Spectroscopic Studies of Iron Based Catalysts during Fischer-Tropsch Synthesis

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Introduction

Knowledge about the structure of active sites and its dynamic changes is essential for a better understanding of catalyst structure-reactivity relationships. For this purpose, *in situ* and *ex situ* Raman experiments were performed to monitor the structural changes of one non-promoted and two promoted iron oxalate catalysts during the hydrogenation of CO_2 to higher hydrocarbons under realistic reaction conditions (340°C, 15 bar).

Experimental

The promoted catalysts $LiFeC_2O_4$ and $KFeC_2O_4$ were prepared through a simple incipient wetness impregnation of commercial iron(II) oxalate dihydrate with an aqueous solution of the alkali-metal carbonate. No catalyst reduction is required, because the catalytically active iron carbides and Fe_3O_4 phases are formed *in situ* under reaction conditions.

The *in situ* Raman spectroscopic experiments were performed at a Renishaw inVia Raman microscope, using a Harrick reaction cell, in which reactions, requiring higher pressures and temperatures can be studied spectroscopically. The cell acts as a fixed bed flow reactor and was connected to a gas supplying system using MFCs and a membrane back pressure regulator. A 633 nm HeNe laser was used with a laser power of 0.17 - 1.7 mW.

Ex situ Raman spectroscopy measurements were carried out using a Horiba iHR550 spectrometer with a 785 nm laser source and a Marqmetrix contact ballprobe with sapphire lens.

Results and discussion

Figure 1 shows relevant *in situ* Raman spectra obtained from the three catalysts collected during high pressure CO_2-FT synthesis, as well as *ex situ* Raman spectra after reaction.

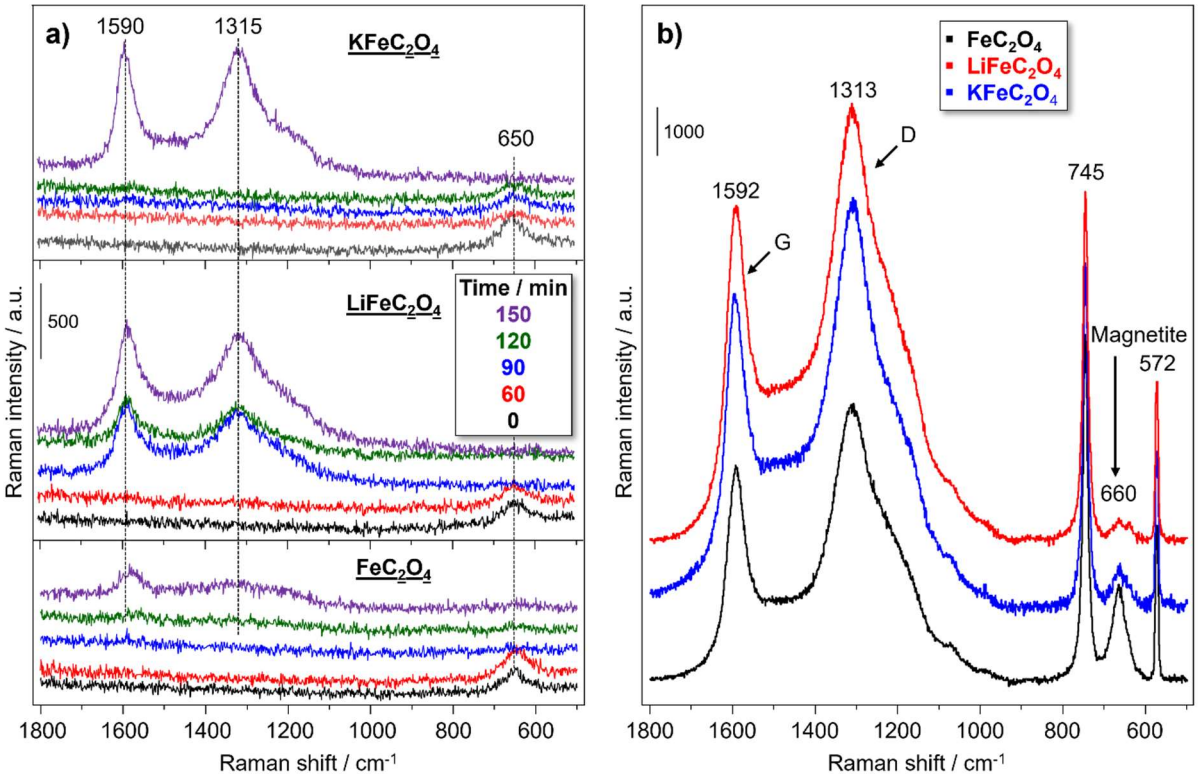

Figure 1. a) In situ Raman spectra of FeC_2O_4, $LiFeC_2O_4$ and $KFeC_2O_4$ during exposure to $CO_2/H_2/He$ (10/30/10; 50 ml/min) at 340°C and 15 bar. b) Ex situ Raman spectra of FeC_2O_4, $LiFeC_2O_4$ and $KFeC_2O_4$ after 180 min reaction.

At reaction temperature and pressure in an inert atmosphere, the formation of Fe_3O_4 (magnetite) phase is clearly visible; however, when reactive gas (CO_2/H_2; 1/3) is added, it vanishes, indicating the synthesis of metallic iron and iron carbides that are not Raman active. Instead, the formation of carbon deposits was identified, by its intensive and characteristic G and D bands. They occur more obviously in doped catalysts than in the undoped one.

While *in situ* Raman spectroscopy provides time-resolved results under reaction conditions, the spectra are of low quality and only represent a single focal point on the surface. However, combining these data with those obtained by *ex situ* Raman spectroscopy, which produces higher-quality spectra of the bulk catalyst, gives a more complete picture of the phase changes of the catalyst during CO_2-FT reaction.

In summary using Raman spectroscopy, the structure evolution in particular the behavior of C deposits under changing gas atmospheres and on different catalysts could be clarified. Further research is needed to determine whether the Raman-observed C deposits are intermediates of the CO_2-FT or simply spectator species, or whether they are even detrimental to the CO_2-FT performance.